

Reference Definitions Reverse Detector Rankings in Streaming Key Estimation

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Abstract. What key is a passage in while the music is still unfolding? An analyst may describe a temporary move to a new key even when the written key signature does not change. A detector that follows the local move and one that retains the broader key can therefore receive opposite scores depending on which annotations are treated as the answer. We study this evaluation problem for causal key estimation from recognized chords, where a system predicts after each event or abstains. On 36 Beethoven performances, we score the same outputs from a responsive short-memory detector and a stable long-memory detector against analyst-declared key contexts and active key-signature collections. We map both references to a common 12-category representation and compare events where both detectors make a claim. The references agree at a mean per-piece rate of 0.65 and reverse the detector ranking: the long-minus-short accuracy difference changes from -0.081 to $+0.080$ (exploratory interaction $+0.160$, CI95 $[+0.118, +0.205]$). Development analyses show corresponding changes in the value of memory and harmonic-function evidence. Neither reference is uniquely correct, and the cross-corpus evidence does not isolate annotation persistence from repertoire. The result is instead methodological: key-estimation scores must identify what their reference annotations mean.

1 Introduction

A passage can move temporarily to another key even while its written key signature remains unchanged. A responsive detector may follow that local move; a more stable detector may retain the broader key. Neither behavior is inherently wrong. Which detector appears better depends on what the annotations used for evaluation are intended to describe.

That problem is easy to overlook because a benchmark score makes its labels look like a neutral answer key. In practice, music datasets encode different notions of key: an analyst’s current interpretive context, a time-bounded tonality region, or the collection implied by a written key signature. These notions differ in meaning and persistence. If a benchmark does not state which one it measures, an accuracy comparison may conflate a detector’s quality with its agreement with a particular annotation practice.

We examine this problem in a streaming setting motivated by a musician playing a MIDI keyboard. After every recognized chord, the system must name a key or withhold judgment using only the music heard so far. This causal boundary makes responsiveness, stability, and abstention visible, but our main

question is more basic: can the definition of the reference answer change which detector wins?

The central experiment uses 36 recorded Beethoven performances with two documented references: analyst-declared key contexts and active notated key signatures. We hold the performed input and both detectors’ predictions fixed, translate all answers to the same 12 diatonic collections, and compare only events where both detectors speak. The references agree on roughly two thirds of events and reverse the ranking of a responsive short-memory configuration and a stable long-memory configuration (Section 6).

The paper contributes this controlled demonstration, supporting analyses of memory and reference persistence, and a causal evaluation protocol that reports how often a detector speaks as well as how often it is correct. The claim is not that one reference is the true key, or that annotation persistence explains every difference between datasets. It is that the meaning of the reference labels is part of the task definition and must accompany a key-estimation score.

2 Related work

Distributional key finding begins with the Krumhansl-Schmuckler probe-tone profiles [1] and revisions such as Temperley’s [2], with corpus-trained profiles improving minor-mode behavior [3]. Bayesian and hidden Markov models add temporal structure [4], [5]. The closest architectural analogue here is justkeydding [6], an HMM over profile emissions for global and local key; neural harmony systems also estimate local key within larger retrospective analyses [7], [8]. Our detector remains deliberately close to this interpretable lineage.

Global key conventionally characterizes a whole song, piece, or movement; local-key estimation emits finer-grained predictions [9], [10]. That output distinction is independent of the information boundary: either task may be causal or retrospective. Nor does a local-key sequence uniquely encode music-theoretical modulations and tonicizations [10]. The reference must therefore be described operationally rather than inferred from the word “local.”

Sequential key induction also has a substantial history. Weber treated tonal understanding as something revised while events unfold [11], [12]; interactive and perceptual studies later modeled key from the evidence available so far [13], [14], [15]. Perceived modulation likewise depends on preceding context [16]. Causality is therefore an established task choice rather than this paper’s novelty.

Reference design is especially consequential across repertoires. The major/minor answer space fits much common-practice analysis but does not exhaust popular tonal organization. The repeated Am-F-C-G axis progression can support both A-minor/Aeolian and C-major hearings [17], and experiments on rotations of the same four chords show that chord identity and metric placement both influence perceived center [18]. Six-based minor and other major/minor mixtures further resist a forced binary reading [19]. Abstention can express insufficient evidence among modeled labels; it cannot represent a confident modal, blues-based, or dual-tonic answer outside the 24-state space.

The reference sources are not interchangeable. When in Rome contains human functional analyses encoded in RomanText [20], [21]; Isophonics provides time-aligned tonality regions [22], [23]; and ASAP provides aligned performances and key-signature metadata [24]. Section 4 traces how each source is transformed and what agreement with it can mean.

3 Task and evaluation protocol

Task. The input is a causal stream of recognized chord events. Each event summarizes the candidate chord identities, sounding voicing, timestamp, and hold duration for one stable sonority. After every event, the detector returns a ranked list of keys with confidence values or explicitly abstains. It never sees future events or revises its earlier output. The modeled answer space contains the 12 major and 12 minor keys; annotations outside that space are unscorable rather than evidence that the detector should have abstained. Section 4 describes how performed notes become stored events without exposing reference labels to the recognizer or detector.

Metrics. The top-ranked claim is scored. Accuracy-like metrics are reported in the selective-prediction frame [25]: coverage (how often the detector makes a claim) and accuracy on claimed events form an inseparable pair, and abstentions are never counted as errors. Exact accuracy is complemented by the MIREX-weighted score, which grants partial credit for musically close misses (fifth 0.5, relative 0.3, parallel 0.2) [26]. At zero coverage, however, accuracy on claims is undefined rather than perfect, so an always-abstaining system cannot win the evaluation.

Temporal behavior gets three measurements. An annotated key change is **matched** when the detector claims the new key before the next change, and its **lag** is counted in events; unmatched changes are reported as censored counts, never averaged into lag. A switch between claims is **spurious** only when the annotation did not change and the new claim is not the annotated key, so a lagged catch-up onto the right key is never penalized twice. Its per-piece distribution includes only pieces with at least one exact local-key reference; raw switches and time to first claim remain behavioral summaries over every piece. Annotated changes require adjacent non-null references, so null-reference regions enter neither the matched-

change count nor the lag denominator. **Time-to-first-claim** counts events before the detector first commits. These counts are long-tailed, so each is summarized by the per-piece median and 90th percentile (p90) rather than a mean. Ambiguity-labeled events (hand-authored fixtures only) accept abstention or any acceptable key.

The coverage-accuracy curve is traced by sweeping the abstention threshold, with the selected operating point marked. This shows whether abstention actually concentrates the detector’s errors rather than merely reducing its output.

Statistics and analysis discipline. Comparisons use the piece, not the event, as the unit of analysis because events within a piece are dependent and long pieces would dominate pooled counts. We report Wilcoxon signed-rank tests and seeded-bootstrap 95% intervals on mean paired differences. Development/test splits and the protocol were fixed before detector selection; all tuning and ablation used development data. The held-out configuration comparison was specified before test execution (Section 7). The dual-reference, segment-persistence, and 2x2 analyses were designed after inspection of the cross-corpus pattern and are exploratory or descriptive. Labels are structurally removed before an event reaches either the chord recognizer or key detector.

4 Data and reproducibility

Evaluation uses **fixtures**: versioned event streams containing the chord recognizer’s candidate rankings, observed voicings, timing metadata, and a parallel reference stream read only by the scorer. Fixtures are generated under a fixed neutral analysis context: the recognizer never sees a reference key, so its tonality-sensitive ranking rules cannot leak the answer into the observations. Results are comparable only within a fixture version because each fixture embeds a specific frozen chord-recognition profile.

Table 1 separates repertoire, observation construction, and reference provenance. The first three sets have frozen development/test splits. The ASAP-WiR overlap is evaluation-only and supports both performed-input analyst contexts and the active ASAP key-signature collection.

The When in Rome subset contains 77 common-practice works by Bach, Brahms, Schubert, Beethoven, and Mozart. Its Roman-Text key field is the analyst’s current interpretive context; an applied-chord figure can mark a tonicization without changing that field. The fixtures contain 514 explicit key changes and 455 events with applied-chord slash figures, so “analyst context” must not be read as “every tonicization.”

The Isophonics set contains 224 popular recordings, mostly the Beatles, with Queen, Zweieck, and Carole King also represented; all 41 held-out tracks are Beatles recordings. The source describes its keys as time-aligned tonality regions, not uniformly whole-song labels, and notes that changes may be omitted. Plain major/minor regions map into the detector’s

corpus	repertoire / input	reference	pieces / events
When in Rome [20]	common-practice / score-derived	analyst-declared key context	77 / 5,207
ASAP [24]	classical piano / performed MIDI	active key-signature collection	60 / 19,546
Isophonics [22], [23]	popular songs / synthesized voicings	time-aligned tonality region	224 / 19,062
ASAP-WiR overlap	Beethoven / performed MIDI	analyst context and signature collection	36 / 10,395

Table 1: Evaluation sets and their reference definitions. Repertoire, observation construction, and reference provenance change together in cross-corpus comparisons.

answer space. Seven tracks contain 568 modal or no-key events outside it; three held-out tracks are wholly modal. They remain in the full split but are excluded from both accuracy and coverage in the 38-track 24-key scoring cohort (Section 7).

ASAP key signatures are not analyst-stated local keys. At each event the signature defines an acceptable major/relative-minor pair; it cannot identify a unique tonic and mode and may remain notationally unchanged across an analytical modulation. For the dual-reference analysis, both signature pairs and major/minor analyst contexts are mapped to the corresponding 12 diatonic- collection classes.

The overlap transfers When in Rome analyst contexts onto ASAP performances of the same Beethoven sonata movements through score-performance downbeat alignment. Each movement’s measure offset is calibrated by content: agreement between sounding pitches and analyst chords must peak sharply at zero when labels are slid by whole measures. Movements without a unique, content- consistent offset are rejected rather than silently aligned.

Performed fixtures are created by offline replay of recorded MIDI with sustain semantics through the production chord analyzer, three-note gate, and event- segmentation core. The detector later consumes the stored events in causal timestamp order. This exercises the core musical-analysis path on performed input; it is not an end-to-end live-user, transport, latency, display, or usability evaluation.

A small hand-authored pop/jazz suite supplies behavioral regressions outside all pooled statistics. The repository contains the open fixtures, split files, evaluation harness, paired-comparison code, dated logs, and frozen result artifacts. Licensed fixtures remain in local builds generated from pinned upstream checkouts. Versioned analysis profiles and detector recipes pin the observation and detector configurations used here independently of mutable software defaults.

5 Two streaming detectors

The detector is deliberately simple enough to inspect. It summarizes the recent pitch classes, gives older chords progressively less weight, and compares that summary with profiles for the 24 major and minor keys [3]. An optional functional term asks whether a chord behaves as expected in each candidate key.

Two fixed configurations organize the principal comparisons. The **long-memory configuration** gives accumulated chord evidence a 30-second half-life and does not use the functional term. The **short-memory configuration** uses a 1-second half-life and adds the functional term at weight 0.1. Both use the same HMM, duration weighting, abstention rule, and same-tonic mode cue. They were selected against different development references and fixed before held-out evaluation; neither is proposed as universally best.

The mode cue uses a recognized tonic chord’s major or minor quality to redistribute evidence only between the major and minor keys sharing that tonic. It preserves the pair’s total evidence, so it cannot favor another tonic or create a key change by itself.

The detector family is a causal HMM over the 24 major and minor keys. It keeps a filtered posterior and updates it by the forward algorithm only; no Viterbi decoding or future context is used. At each event, the update is:

```

for each pitch class p:
    histogram[p] =
        sum over prior events i of:
            duration_i
            * 2^(-age_i / half_life)
            * indicator(p in voicing_i)

for each candidate key:
    profile_score[key] =
        correlation(histogram, key_profile[key])

emission = softmax(profile_score / temperature)
prior =
    transpose(transition) * previous_posterior
posterior = normalize(emission * prior)

```

Multiplication in the last line is elementwise. In words: the profile scores become an event-level likelihood over keys, the previous belief is advanced through the transition model to form a prior, and the two are combined and renormalized into the filtered belief after the current event.

In the long-memory configuration, `key_profile` is the Albrecht-Shanahan major/minor profile set [3]. Each event’s contribution to the histogram is multiplied by its duration, and `half_life = 30 s` means that contribution halves after 30 seconds. The softmax temperature is 0.25, sharpening the profile-correlation scores into a more selective likelihood. The

transition matrix assigns 0.9 probability to remaining in the same key; the remaining mass is distributed over other keys with decay by circle-of-fifths signature distance. The frozen detector withholds its first two events. From the third event onward, it reports the posterior’s top key and probability only when the margin between the top two probabilities is at least 0.3; otherwise it abstains. The three-event warm-up and margin floor are independent claim gates.

The decayed histogram and the HMM transition model both preserve information, but they play different roles: the histogram defines how much recent musical evidence forms the current observation, while the transition model discourages the hidden key from changing at every chord. The experiments below vary the first form of memory while keeping the second fixed.

6 Reference-dependent evaluation

6.1 Fixed outputs, different references

The primary analysis asks whether two documented reference definitions select the same detector on unchanged material. It uses all 36 ASAP-WiR Beethoven performances and the fixed long- and short-memory claim streams. Analyst keys, key signatures, and detector claims are mapped to 12 diatonic-collection classes: a minor key shares a class with its relative major. This gives both references the same cardinality while deliberately discarding major/minor identity. The primary event mask contains the 8,160 events on which both configurations claim and both references exist.

reference	long	short	long - short, CI95
analyst key context	0.581	0.661	-0.081, [-0.137, -0.026]
key-signature collection	0.626	0.547	+0.080, [+0.029, +0.128]

Table 2: Exact accuracy on common claims under two references, macro-averaged by performance in a shared 12-class representation. Intervals are exploratory paired bootstrap intervals. Neither detector output changes.

Mean per-piece reference agreement is only 0.6465. Under the analyst context, the short-memory configuration leads; under the active key-signature collection, the long-memory configuration leads (Table 2). The piece-level difference of differences is +0.1602, with exploratory CI95 [+0.1184, +0.2046]; 31 of 36 pieces have a positive interaction. The same reversal appears when each configuration is evaluated on its own claims, so the common-event restriction does not create it.

This is sensitivity to the reference definition, not a contest to identify the one correct reference. An analyst’s current key context and the collection implied by a notated signature answer different valid questions. The result is also not a pure timescale manipulation: source, semantics, and persistence all change with the reference. What the design establishes directly is that the reference definition can reverse model selection while performances and predictions remain fixed.

6.2 Memory and function across references

The two fixed configurations bundle memory and functional evidence, so their reversal alone cannot say which ingredient matters. An exploratory 2x2 grid, specified before those cells were run, crosses half-life {1, 30} seconds with functional blend {0, 0.1} on both development sets while fixing every other setting (Table 3). The corpora still differ in repertoire, observation construction, and reference practice.

reference	func.	1 s	30 s
WiR contexts	0.0	0.546 / 0.680	0.434 / 0.784
WiR contexts	0.1	0.601 / 0.764	0.514 / 0.824
Isophonics regions	0.0	0.736 / 0.791	0.775 / 0.921
Isophonics regions	0.1	0.629 / 0.815	0.713 / 0.926

Table 3: Development cell means when memory and functional evidence are crossed independently; detector cells report exact accuracy / coverage. Exact accuracy is conditional on claims; WiR is When in Rome.

On When in Rome, 30-second minus 1-second exact accuracy is -0.104 without functional evidence and -0.089 with it; both exploratory intervals exclude zero. On Isophonics the corresponding effects are +0.040 (CI95 [-0.0001, +0.0786]) and +0.084 (CI95 [+0.041, +0.129]). Function helps When in Rome at either memory (+0.063 and +0.081) and harms Isophonics at either memory (-0.107 and -0.062); all four functional-effect intervals exclude zero. Longer memory raises coverage in every pair, while functional evidence raises or preserves it, so the opposite exact-accuracy effects are not produced by opposite abstention directions.

The familiar smoothing explanation correctly anticipates that persistent references may reward longer evidence windows. It does not by itself predict the magnitude of the reference disagreement in Table 2 or the sign change of a modeling ingredient in Table 3. Those are the evaluation and model-selection consequences established here.

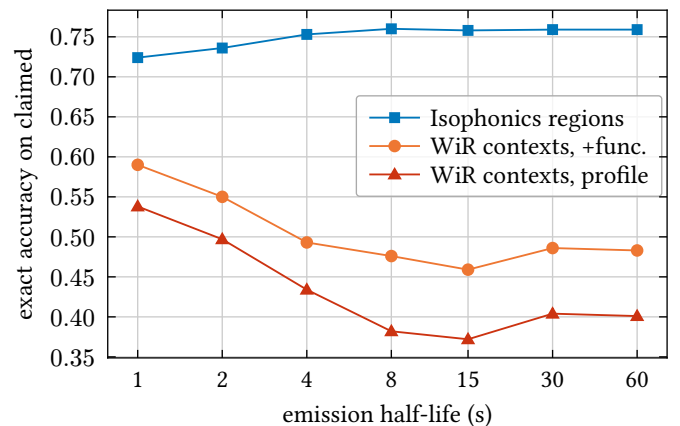


Figure 1: Descriptive development sweep. When in Rome exact accuracy declines strongly to 15 s and partially rebounds; Isophonics rises to a broad plateau from 8 s. Coverage rises with memory in all three series. No one memory setting is favored by both reference regimes.

Figure 1 sweeps the emission-memory half-life from 1 to 60 seconds on both development corpora. The plotted series confirm the endpoint pattern without being strictly monotonic: both When in Rome curves reach a minimum at 15 s and partially rebound, while Isophonics reaches a plateau by 8 s. The 2x2 endpoint contrasts above supply paired uncertainty; the full inspected sweep is descriptive.

6.3 Reference persistence on performed input

The overlap also permits a within-reference descriptive analysis. Each event keeps its projected analyst context; the analysis only restricts eligibility to contexts that persist for at least 0, 12, 20, or 32 score measures. It therefore changes the event subset, not the labels assigned to identical events. The thresholds and pooled direction had already been inspected, so Table 4 uses piece-level summaries and descriptive intervals rather than a new confirmatory threshold claim.

min. span	n	long	short	diff. / CI95
0	36	0.895 / 0.502	0.846 / 0.605	-0.103, [-0.158, -0.046]
12	36	0.909 / 0.600	0.863 / 0.628	-0.028, [-0.084, +0.029]
20	35	0.907 / 0.625	0.864 / 0.624	+0.001, [-0.067, +0.069]
32	30	0.913 / 0.691	0.875 / 0.644	+0.046, [-0.030, +0.120]

Table 4: Piece-level ASAP-WiR results as evaluation is restricted to longer- persistence analyst-key segments. Detector cells report coverage / exact accuracy; spans are score measures. Intervals are descriptive paired bootstrap intervals. Contributing pieces and coverage change with the filter.

Relative accuracy shifts progressively from the short-memory configuration toward the long-memory configuration as the minimum segment span rises. The same trend remains on the common-claim events: long-minus-short exact accuracy moves from -0.093 to +0.033. Only the all-event short-memory advantage has an interval excluding zero; the later near tie and long-memory lead are descriptive. This supports an association between reference persistence and configuration fit, not a sharp 12-measure crossover or proof that persistence alone explains the cross-corpus results.

Together, the three analyses establish different levels of evidence. The dual-reference result directly holds predictions fixed. The development grid shows that bundling does not create the memory and functional patterns. The persistence analysis connects those patterns to a measured temporal property within one performed-input analyst reference. Repertoire

and observation construction remain alternative contributors to the broader When in Rome versus Isophonics contrast.

6.4 Abstention behavior

Sweeping the posterior-margin floor traces the selective-prediction curve in Figure 2. Higher thresholds reduce coverage while raising accuracy on the remaining claims. An all-abstaining detector would have zero coverage and no claimed-event accuracy; it cannot score as a useful system. The margin floor of 0.3 was selected on development data before held-out execution. That point is evaluated as a coverage-accuracy pair, with stability and latency reported separately.

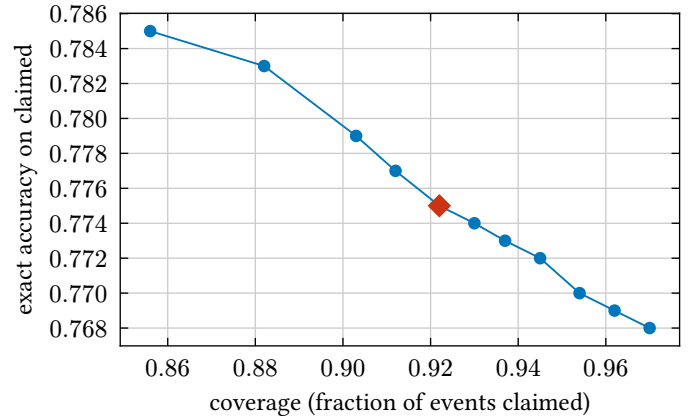


Figure 2: Selective-prediction behavior of the fixed long-memory configuration on the Isophonics development split, swept over the posterior-margin floor (0 to 0.6). Moving left raises the margin required to speak, so coverage falls; moving up means the remaining claims are more often correct. The marked point is the evaluated operating point (floor 0.3).

7 Held-out evaluation

The held-out manifest was declared before execution. It evaluates the long- memory configuration on all three splits, the short-memory configuration on the two splits whose references distinguish major from minor, and three music21 [27] profile-correlation analyzers on Isophonics. The resulting claim streams are fixed, and every cohort restriction reported below depends only on the reference labels.

split	scorable / total	coverage	exact
Isophonics	38 / 41	0.884	0.732
WiR	18 / 18	0.811	0.587
ASAP	10 / 10	0.830	0.683

Table 5: The fixed long-memory configuration on the three held-out splits. Three wholly modal Isophonics tracks have no event in the 24-key answer space and are excluded from both accuracy and coverage, while remaining in the frozen split. ASAP is scored against acceptable major/minor realizations of its key signatures.

The long-memory configuration retains substantial coverage across all three inputs (Table 5). Development-to-test changes differ by corpus (Isophonics exact falls from 0.775 to 0.732; When in Rome rises from 0.434 to 0.587), so these small splits

support a generalization check rather than a precise estimate of deployment performance.

The predeclared ordering reverses across references.

Against When in Rome analyst contexts, the short-memory configuration exceeds the long-memory configuration (0.649 versus 0.587 exact; paired difference +0.062, CI95 [+0.004, +0.121], $p = 0.047$; 16 paired pieces), at lower coverage (0.745 versus 0.811). Against Isophonics region labels, the long-memory configuration exceeds the short-memory configuration (0.732 versus 0.556; +0.175, CI95 [+0.040, +0.315], $p = 0.039$; 38 pieces), at higher coverage (0.884 versus 0.793). These are two within-reference tests, not a formal interaction test, and the configurations bundle memory with functional weighting. They are therefore a generalization check consistent with the fixed-output interaction, not its substitute.

Descriptive external reference points. Table 6 reports three classic offline whole-piece profile-correlation analyzers from music21 on the held-out Isophonics songs.

system	coverage	exact	MIREX
fixed long configuration	0.884	0.732	0.782
Temperley-Kostka-Payne [4]	1.00	0.637	0.740
Krumhansl-Schmuckler [1]	1.00	0.624	0.726
Aarden-Essen [28]	1.00	0.558	0.690

Table 6: Descriptive held-out Isophonics reference points for the causal, abstaining long-memory configuration and three offline whole-piece profile-correlation analyzers.

The long-memory configuration has the highest point estimate, while Temperley-Kostka-Payne is the strongest whole-piece reference point. The Krumhansl-Schmuckler contrast is +0.108, but its interval spans zero ($[-0.008, +0.228]$, $p = 0.25$), and no superiority or equivalence test was specified. More importantly, the systems answer different questions: each offline analyzer reads the entire song and returns one key, whereas the causal configuration may change or abstain after each chord. The table supplies familiar context, not evidence that the streaming system matches or surpasses the offline key-estimation literature.

8 Limitations

The primary dual-reference result is deliberately narrow: two fixed detector configurations, 36 Beethoven performances, their common claimed events, and a shared 12-class diatonic-collection representation derived from 24-key outputs. It isolates the reference definition while holding input and output fixed, but it does not establish the size or direction of the effect for other repertoires, answer spaces, or detectors. The dual-reference and 2x2 analyses were designed after inspection of the cross-corpus pattern and are exploratory; their intervals quantify uncertainty but do not convert them into preregistered confirmatory tests.

Neither reference is ground truth. Analyst-declared contexts and active notated key-signature collections encode different musical questions, and either may be debatable in ambiguous passages. The broader cross-corpus contrasts also change repertoire, observation construction, and reference provenance together. Reference persistence is one measured contributor, not a complete causal explanation.

The detector’s answer space contains only 12 major and 12 minor keys. Modal, blues, mode-mixture, and tonic-ambiguous loops can fall outside it or admit multiple reasonable readings; three wholly modal Isophonics test tracks are consequently excluded from 24-state accuracy and coverage. They remain in the frozen split as a separate behavioral audit. The recognizer is also part of the measurement chain: fixtures embed its chord rankings, so recognizer changes require regenerating fixtures before comparison.

“Streaming” here means causal evaluation during offline replay of recorded performed MIDI. The study does not measure wall-clock latency, transport failures, interface behavior, or musician judgments in a live session. The 30-second and 1-second configurations are fixed experimental recipes, not an end-to-end application evaluation.

Held-out splits contain only 10 to 41 pieces. The offline comparison is limited to three classic profile-correlation analyzers with different information and output forms. In particular, justkeydding [6] did not build reproducibly in our environment, so no claim is made against it or newer score-based systems. Corpus licensing further prevents redistribution of two gated fixture sets; the project instead records pins, derived facts, splits, commands, and evaluation artifacts.

9 Conclusion

A detector can appear better or worse depending on what the reference calls the key. On the same Beethoven performances and the same detector outputs, analyst-declared key contexts favor the short-memory configuration by 0.081, while notated key-signature collections favor the long-memory configuration by 0.080. The resulting paired interaction is +0.160, with an interval excluding zero.

The supporting analyses clarify rather than erase that result. Memory and functional evidence have opposing effects under the two development reference regimes; longer-persistence analyst contexts progressively favor longer memory; and the configuration reversal reappears on held-out data. Repertoire and observation construction still differ across the broader corpus comparison, so the fixed-output dual-reference experiment remains the cleanest evidence.

The study also provides a reproducible protocol for a causal detector that may abstain, reporting coverage, accuracy on claims, change lag, and spurious switching together. It does not establish offline parity or end-to-end live usability. Its narrower conclusion is more fundamental: a key-estimation

score cannot be interpreted without knowing what kind of key its reference encodes. Benchmarks should therefore state the reference’s source, temporal granularity, and treatment of tonicization, signature, mode, and ambiguity.

Future work should repeat the fixed-output design across repertoires and detector families, expand beyond major and minor keys, and evaluate the causal system with musicians in live interaction.

Competing Interests

The author has no competing interests to declare.

Reproducibility: protocol, logs, corpus pins, splits, harness, and held-out evaluation artifacts live under `research/whatkey/` in the WhatChord repository. Every number traces to a dated log entry and a report with its generating command.

<https://github.com/EarthmanMuons/whatchord>

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